

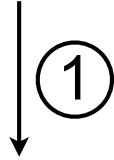
Discrete time

(Geometric growth)

Continuous time

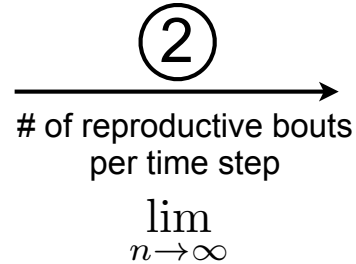
(Exponential growth)

Difference eqn. $N_{t+1} = (1 + r_d)N_t = \lambda N_t$



Projections

$$N_t = \lambda^t N_0$$



Differential eqn.

$N_t = N_0 e^{rt}$

③ ↓ ④ ↑

$\lim_{\Delta t \rightarrow 0}$

$$\frac{dN}{dt} = rN$$